

COMP3320 Week 1, Lecture 1

Rhys Hawkins

Semester 2, 2024

Dr Rhys Hawkins

Course Convener, Lecture

Ryan Stocks, Elise Palethorpe

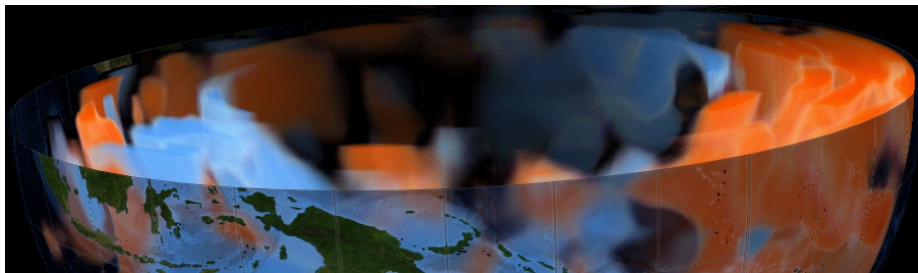
Tutors

About the Course

How to get the most out of modern CPUs for Scientific Computing

- Accuracy (Floating Point)
- Measurement
- Efficiency (operations and memory access)

My Background



- Seismic tomography based on waveform simulation
- Large $> 100\text{M}$ dof simulations
- Regional seismic hazard simulations

Contact

- Preferred general communication is through Ed Discussions
- Expect an invite in Week 1 or 2.
- comp3320@anu.edu.au for administrative type questions

Textbooks

The following text book provides good coverage for a large part of the course.

- *Introduction to High Performance Computing for Scientists and Engineers*, Georg Hager and Gehard Wellein, CRC Press, ISBN 9781439811924.

Furthermore, some lectures in the second part of the course are covered by

- *Computer Systems A Programmer's Perspective, Third Edition*, Randal E. Bryant and David R. O'Hallaron, Pearson Education Heg USA, ISBN 9781292101767. We will cover Chapter 12 – Concurrent Programming.
- *Using OpenMP Portable Shared Memory Parallel Programming*, Barbara Chapman, Gabriele Jost and Rudd van der Pas, MIT Press, ISBN 9780262533027.

Other references are detailed on the course Wattle site. We will also provide additional material on a few topics.

Lectures

Details in Course Schedule on Wattle

Lecture	Time	Location
1	Mon 11:00 – 12:00	Psychology Building 39 G05
2 & 3	Tue 14:00 – 16:00	Psychology Building 39 G05

- Lecture Slides hopefully available before lecture (same day)
- Recording uploaded to Wattle (hopefully) same day

Labs

Details in Course Schedule on Wattle

Lab	Time	Location
1	Tue 12:00 – 13:30	CSIT Building 108 N109
2	Fri 14:00 – 15:30	CSIT Building 108 N109

- **No labs this week**
- Held in weeks 2, 3, 5, 8, 9, 11
- **Attendance not compulsory but Labs contain new and examinable material**

Drop-in Sessions

Drop-in	Time	Location
1	Thu 12:00 – 13:00	CSIT Building 108 N115/N116

- Held in weeks 4 and 10
- Designed for students struggling with aspects of their assignments

Enrolment for NCI (National Computational Infrastructure)

Create Account

- See instructions 'Obtaining an NCI account' in Wattle
- Please complete by end of the week (before first lab at the latest).
- <https://my.nci.org.au/mancini/login>
- After following the instructions, you will receive an email containing a link to confirm your registration.

Gadi User Guide (Need Account First)

- <https://opus.nci.org.au/display/Help/Preparing+for+Gadi>
- Basic information on the use of the machine
- Includes a link to an FAQ page.
- You should take a look at this and use it for reference throughout this course.

Assignments

Assignment will take the form of two projects that will run before and after the mid-semester break.

- Project 1: Molecular Dynamics Simulation, worth 20 marks, at the end of the mid-semester break.
- Project 2: Cloth Simulation, worth 20 marks, due at the end of week 12.

Detailed project specification will be placed on the COMP3320/6464 Wattle site.

Assessment

The total workload for this (6 units) course is around 150 hours (around 12 hours per semester week). For semester 2 2020, the following assessment protocol is proposed:

- **Assignments – 40% of final mark**

- There will be two assignments, each worth 20% of the final course mark (in total 40% of the final course mark).
- These assignments will take the form of programming tasks and/or answering practical and theoretical questions relevant to the course content.

- **Mid-Semester Test – 20% of final mark**

- There will be a 60 minute exam held in week 4 this will be worth 20% of your final course mark.
- If you miss the mid-semester exam for a legitimate reason (e.g. medical) you will be awarded zero and your final course make will be derived by scaling your final exam mark to represent 60%.

- **Final Exam – 40% of final mark**

- This will be a 3 hour written exam held at the end of the semester during the normal examination period.
- This exam is worth 40% of the final course mark.

Final Course Mark

- To pass this course you must score at least 50 out of a 100 marks, and you must score at least 45% in the final written examination.
- A supplementary exam will be offered only to those students who have a final mark of at least 45 out of 100, but less than 50 out of 100.
- The final mark will be computed as follows

```
Final_Mark = Project_1_Mark+Project_2_Mark+Mid_Semester_Mark+Final_Exam_Mark;  
if ((Mid_Semester_Mark + Final_Exam_Mark)/60 < 27/60) {  
    Final_Mark = Min(44, Final_Mark);  
}
```

- **From the date that the project marks are released electronically, you will have a period of two weeks in which to question your mark. After this period your mark will be final.**
- Release of your project marks will be announced on the Wattle site and noted in lectures.

Extensions and Late Penalties for Assignments

In general, there will be **no extensions for assignments**.

- The exception being for illness serious enough to keep you in bed, supported by a medical certificate. Other similarly unforeseeable and serious circumstances may be considered (if similarly verified).
- Work and sporting commitments are not normally sufficient grounds.
- Extensions are managed through centralized process and you apply via a form (link below and on Wattle)
- Extenuating Circumstances apply for exams and longer absences

Links to Extension and Extenuating Circumstances

<https://cecc.anu.edu.au/students/continuing/assessments/>

Plagiarism

- No group work is permitted for any part of the assessment. We do encourage you to discuss your work in the labs and lectures, but we expect you to do the submittable work by yourself.
- We do take plagiarism seriously! Your work must be genuine and original. Please make sure you read the following page:
<https://www.anu.edu.au/students/academic-skills/academic-integrity/plagiarism>
- Files you have found on the Internet should not be submitted as your work; but your documents may include URL links to external documents. Always reference work.
- Avoid copying or closely paraphrasing material from documents that you find, including those of your friends or textbooks.

Plagiarism

- If you feel it necessary to include material from some other document, then it should be clearly identified as such, identifiable as quoted material by layout or quotation marks, and proper attribution made.
- Be aware that automated tools to detect plagiarism exist and might be used (for example Glatt or JPlag)
- Receiving significant assistance or, worse, paying someone else to write your assignments for you is academic misconduct!

At ANU, plagiarism is dealt with in a variety of ways depending on the circumstances of the 'case'. **Deliberate academic misconduct is a very serious situation and may lead to expulsion from the university.**

Generative AI Tools

- AI Generative Tools such as ChatGPT are **NOT** allowed in the coding or written parts of the assignments.
- The use of AI Generative Tools to complete either assignment is considered a case of academic misconduct.

Special Exam Arrangements (SEAs)

- To support students with disability, health condition, carers
- Deadlines to renew/request SEAs
 - August 4th for Mid Semester Tests and Assignments
 - October 11th for Final Exam
- Contact access.inclusion@anu.edu.au and us to arrange a plan
- <http://www.anu.edu.au/students/contacts/access-inclusion>

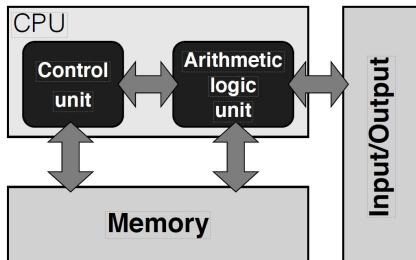
Course Representatives

- At least one student from each course 3320/6464
- If no one volunteers, we have to choose someone
- Course representatives provide constructive feedback to course conveners from the whole cohort
- **More info:** <https://cecs.anu.edu.au/current-students/opportunities-and-projects/professional-development/cecs-class-representatives>

Course Schedule

Week Starts	Lectures	Assignment	Laboratory Content
22nd July	Background Admin, Von Neumann Measuring Performance		
29th July	Floating Point Assignment Intro.	Assignment 1 Start Molecular Dynamics (MD)	L1: Getting Started Performance Benchmarking
5th August	CPU and Memory Architecture		L2: Hardware Architecture and Code Performance
12th August	Mid-Semester Test		Drop-in: Assignment 1
19th August	Compiler Optimisations Optimisations		L3: Loops and SIMD
26th August	Data Access Patterns		
Mid Semester Break (Sept 5th - Sept 19th)			
16th September	Why Parallel Computers? Pthreads	Assignment 2 start: Cloth Simulation	
23rd September	Pthreads Memory Model OpenMP: Intro		L4: OpenMP
30th September	OpenMP: Fundamentals and Worksharing		L5: Advanced OpenMP
7th October	OpenMP: Synchronisation and Other Features Parallel Computer Hardware		Drop-in: Assignment 2
14th October	OpenMP: SIMD, Parallel Performance and Roofline		L6: OpenMP SIMD
21st October	Course Review and Exam Preparation		
Exam Period (Oct 31st - Nov 18th)			

Von Neumann Architecture



CPU

- Arithmetic/Logic unit – Does computations
- Control Unit – Reads and Executes instructions
- Interfaces

Input/Output (I/O)

- Communications with Users

Memory

- Stores Data
- Stores Instructions (as numbers = data)

What sets von Neumann Architecture apart From earlier designs?

Basic Operation of a Von Neumann Machine

Loop forever

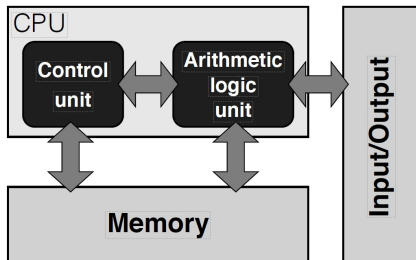
- 1 Load next instruction from **PC** (Program Counter) address in **Memory**
- 2 Decode instruction
- 3 Load operand(s) from **Memory** to registers
- 4 Perform Operation
- 5 Write result from register to **Memory**
- 6 Move **PC** to next instruction

- Each step takes a fixed number of cycles
- Operations typically act on one or more registers (sometimes special purpose)
- Values from Memory need to be transferred to/from registers before/after operations.
- Performance is impacted by
 - Memory read/write operations
 - Operations (some more costly than others)
- CPU does no useful work when
 - it has to wait for memory to be read/written
 - operations to complete
- Results in no-ops

Von Neumann Advantages

- Control Unit retrieves data and instruction in same way from same memory.
- Results in simplified electronics and therefore cheaper.
- Software development simpler (in some ways)
- No arbitrary limitations on memory (instruction versus data memory)

Von Neumann Limitations



von Neumann Bottleneck

- Instructions and data must be continuously fed to the control unit
- Speed of the memory interface limits performance
- Inherently sequential/serial: Single Instruction Single Data (SISD)

We will study architectural optimisations and programming techniques to mitigate these limitations

Von Neumann Bottle Neck

A quote from an article in 1996

An anecdote: in a recent database benchmark study using TPC-C, both 200-MHz Pentium Pro and 400-MHz 21164 Alpha systems were measured at 4.2–4.5 CPU cycles per instruction retired. In other words, three out of every four CPU cycles retired zero instructions; most were spent waiting for memory.

from Bell et al. (1996). *Architects look to processors of future*. Microprocessor Report, Microdesign Resources, 10(10).

Von Neumann Alternatives

- Vector Processors (rare these days)
- SIMD at the hardware level
- See Chapter 1, Section 6 of Text Book
- GPUs are similar
- General purpose CPUs (von Neumann) are dominant

High Performance Computing

Achieving High Performance requires

- Good Algorithms
- Efficient Memory Access
- Optimising Operations (Floating Point)
- Correctness

In the first part of this course we focus on primarily the last three.

Prelude

```
int i;  
float A[N], B[N], C[N];  
for (i = 0; i < N; i++) {  
    C[i] = C[i] + A[i]*B[i];  
}
```

- We will typically analyse code snippets such as this loop
- Break down the inner loop into loads, stores, operations and their order
- Rudimentary C needed, but it is more about understanding how this translates at a lower level into memory access and floating point operations.

Prelude

```
int i;  
float A[N], B[N], C[N];  
for (i = 0; i < N; i++) {  
    C[i] = C[i] + A[i]*B[i];  
}
```

Pseudo Assembly Language

- 1 Load A[i] to register (read)
- 2 Load B[i] to register (read)
- 3 floating point multiplication
- 4 Load C[i] to register (read)
- 5 floating point add
- 6 Store register to C[i] (write)

Prelude: Summary

```
int i;
float A[N], B[N], C[N];
for (i = 0; i < N; i++) {
    C[i] = C[i] + A[i]*B[i];
}
```

- Understanding the relationship between C expressions and low level instructions
 - loads
 - stores
 - floating point operations
- Fortunately in C (and Fortran) this is generally straight forward
- Performance analysis and optimization in terms of
 - Memory operations (Loads and Stores and interaction with Cache)
 - Floating point operations

Why we care about serial performance

Our ability to improve performance by parallel processing is dictated by serial performance

$$P = \frac{1}{s + \frac{p}{N}} \quad (1)$$

- Amdahl's law
- P is the parallel performance
- s is the serial component of an algorithm
- p is the parallel component
- Reducing s and p are fundamentally serial optimization
- Parallel scalability is limited by serial parts of code

What you need to do

- See Week 1 Checklist on Wattle
- Course representatives
- Ed Discussions
- Lab Enrolment
- NCI Account: Please create account before first lab.